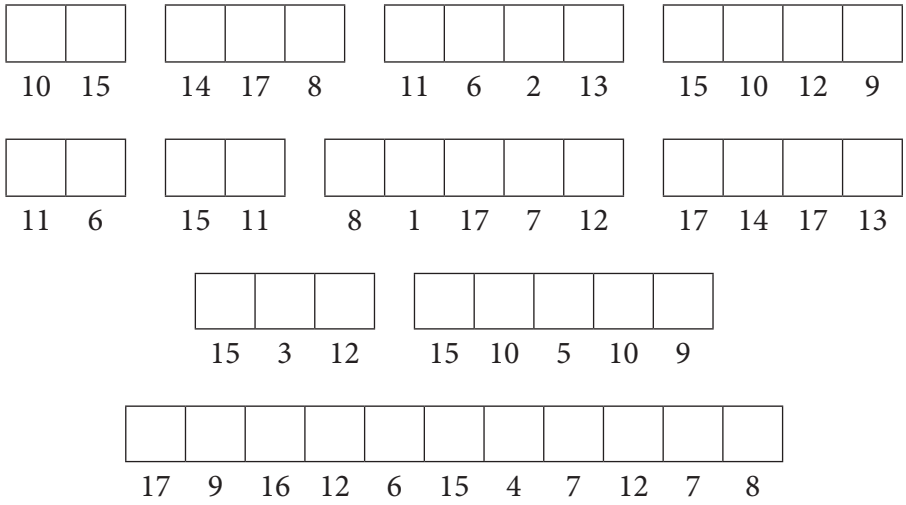
WORKSHEET

Chapter 2: Polar complex number operations

When a resolute young fellow steps up to the great bully, the world, and takes him boldly by the beard, he is often surprised to find it comes off in his hand, and that …



Ralph Waldo Emerson

Questions

1 $\left[2\ \text{cis}\left(\frac{\pi}{5}\right)\right]\left[4\ \text{cis}\left(\frac{\pi}{3}\right)\right]$

2 $\left[\sqrt{50}\ \text{cis}\left(\frac{3\pi}{10}\right)\right]\left[\sqrt{2}\ \text{cis}\left(\frac{\pi}{2}\right)\right]$

3 $\dfrac{9\ \text{cis}(\pi)}{\left[3\ \text{cis}\left(\frac{2\pi}{7}\right)\right]}$

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4 $\dfrac{\sqrt{48}\ \text{cis}\left(\frac{3\pi}{4}\right)}{\sqrt{3}\ \text{cis}\left(\frac{2\pi}{3}\right)}$

5 $\left[7\ \text{cis}\left(-\frac{\pi}{5}\right)\right]\left[5\ \text{cis}\left(\frac{\pi}{2}\right)\right]$

6 $\dfrac{6\ \text{cis}(140°)}{3\ \text{cis}(85°)}$

7 $\left[\frac{3}{4}\ \text{cis}(70°)\right]\left[8\ \text{cis}(-120°)\right]$

8 $\dfrac{0.9\ \text{cis}\left(\frac{\pi}{3}\right)}{0.15\ \text{cis}\left(\frac{\pi}{4}\right)}$

Change the complex numbers below into polar form:

9 $-5 + 5i$

10 $\sqrt{3} + 3i$

Hence, calculate in polar form:

11 $(-5 + 5i)\ (\sqrt{3} + 3i)$

12 $\dfrac{-5 + 5i}{\sqrt{3} + 3i}$

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Change the complex numbers below into polar form:

13 $2 - 2i$

14 $-7 + 7\sqrt{3}\,i$

Hence, calculate in polar form:

15 $(2 - 2i)(-7 + 7\sqrt{3}\,i)$

16 $\dfrac{2 - 2i}{-7 + 7\sqrt{3}i}$

Calculate:

17 $\left[7\sqrt{2}\ \text{cis}(\pi)\right](6 + 6i)$

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Solutions

A $84\ \text{cis}\left(\frac{3\pi}{4}\right)$

C $8\ \text{cis}\left(\frac{8\pi}{15}\right)$

D $5\sqrt{2}\ \text{cis}\left(\frac{3\pi}{4}\right)$

E $\frac{5\sqrt{6}}{6}\ \text{cis}\left(\frac{5\pi}{12}\right)$

H $3\ \text{cis}\left(\frac{5\pi}{7}\right)$

I $2\sqrt{3}\ \text{cis}\left(\frac{\pi}{3}\right)$

L $10\ \text{cis}\left(\frac{4\pi}{5}\right)$

M $35\ \text{cis}\left(\frac{3\pi}{10}\right)$

N $2\ \text{cis}(55°)$

O $10\sqrt{6}\ \text{cis}\left(-\frac{11\pi}{12}\right)$

R $6\ \text{cis}(-50°)$

S $6\ \text{cis}\left(\frac{\pi}{12}\right)$

T $28\sqrt{2}\ \text{cis}\left(\frac{5\pi}{12}\right)$

U $4\ \text{cis}\left(\frac{\pi}{12}\right)$

V $\frac{\sqrt{2}}{7}\ \text{cis}\left(-\frac{11\pi}{12}\right)$

W $14\ \text{cis}\left(\frac{2\pi}{3}\right)$

Y $2\sqrt{2}\ \text{cis}\left(-\frac{\pi}{4}\right)$

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